Math 115A Spring 2026 Discussion 3 Handout

Ben Slater

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1 Introduction

Today we'll work through some exercises about vector spaces and subspaces.

2 Fields and Vector Spaces

First, some exercises about vector spaces and fields. (Mostly repeated from Worksheet 2).

Let's prove some of the statements about vector spaces left unproven in the notes.

Exercise: Let V be a vector space. Show that the zero element 0_V is unique.

Exercise: Let V be a vector space over the field F. Show that for $a \in F$ and $v \in V$, we have $av = 0_V$ iff $a = 0_F$ or $v = 0_V$.

A common sort of problem is to show that a given set has the structure of a vector space. Once you get a feel for these, they're mostly all the same, but it never hurts to have an example or two.

Exercise: Show that $\mathbb{C}$ is a vector space over $\mathbb{R}$ (using the ordinary addition and multiplication operations).

Exercise: More generally, suppose that F is a field and $E \subseteq F$ is a subfield (i.e. a subset of F which is closed under addition and multiplication, and which contains identities and inverses). Show that F is a vector space over E.

Exercise: Let V be the set of functions $f : \mathbb{R} \to \mathbb{R}$ such that $\int_{-\infty}^{\infty} f(x)dx$ is finite. Show that V is a vector space over $\mathbb{R}$, with the addition and scalar operations you would expect. You may assume any facts from calculus you might wish to use.[1]

Exercise: Usually, *magic square* is a 3×3 grid of distinct positive integers such that the sum of the elements along each row, column, and diagonal is always the same; let M denote the set of magic squares. Let M_G denote the set of "generalized magic squares," which are 3×3 grids of real numbers such

[1] This vector space is very closely related to what is usually called $L^1(\mathbb{R})$, although in order to have the "true" $L^1(\mathbb{R})$ you need a slightly different definition of the integral. A related vector space is $L^2(\mathbb{R})$, which consists of functions f such that $\int_{-\infty}^{\infty}(f(x))^2dx < \infty$. You can try proving that this is a vector space as well, but it's much trickier.

that the sum of the elements along each row, column, and diagonal is the same. Show that $M_G$is a vector space over $\mathbb{R}$. (As a challenge, show that M is not).

As a reminder, the axioms which any vector space V (over a field F) must satisfy are:

- (VS1): for all $x, y \in V$, $x + y = y + x$.
- (VS2): for all $x, y, z \in V$, $(x + y) + z = x + (y + z)$
- (VS3): There exists $0_V \in V$ such that for all $x \in V$, $0_V + x = x + 0_V = x$.
- (VS4): For each $x \in V$, there exists a corresponding $y \in V$ such that $x + y = 0_V$.[2]
- (VS5): For all $x \in V$, $1_F \cdot x = x$.
- (VS6): For all $a, b \in F$ and all $x \in V$, $(ab)x = a(bx)$
- (VS7): For all $a \in F$ and all $x, y \in V$, $a(x + y) = ax + ay$.
- (VS8): For all $a, b \in F$ and all $x \in V$, $(a + b)x = ax + bx$.

Remember that you must check each of these properties to establish that a given set is indeed a vector space. For a slightly funkier example, consider this:

Exercise: Consider the set X of positive real numbers, with the operation $a \oplus b = ab$; we have a "scalar multiplication" $\otimes$ defined as follows: for $a \in \mathbb{R}$, $b \in X$, set $a \otimes b = b^a$. Show that X, together with the given operations, is a vector space over $\mathbb{R}$. (Note that you'll need to find 0_X yourself).

Challenge: Consider a set A, and let V be the set of subsets of A. Find suitable "addition" and "scalar multiplication" operations so that V is a vector space over $\mathbb{F}_2$, the field of two elements mentioned in the notes.

Finally, when trying to wrap your head around a mathematical definition, the non-examples are often as instructive as the examples, if not more.

Exercise: Consider the set $X = \mathbb{R}^2$ with the "addition" operation $\boxplus$ given by $(a, b) \boxplus (c, d) = (a + c + 1, b + d + 1)$, and with ordinary scalar multiplication, that is, $k(a, b) = (ka, kb)$. Show that these operations on X do *not* give it the structure of a vector space over $\mathbb{R}$.

Exercise (Challenge): Show that $\mathbb{Z}$, with the usual addition, is not a vector space over $\mathbb{R}$ with *any* scalar multiplication. (Hint: you can't just assume that the scalar multiplication is the "obvious one;" still, the axioms of a vector space place certain restrictions on what this multiplication can be, particularly multiplication by a rational number).

[2] As a warning, this is **not** the same claim as "$\exists y \in V \forall x \in V$ such that $x + y = 0_V$; as an exercise, try to figure out the difference for yourself.

3 Subspaces

Let V be a vector space over a field F. Recall that a subspace of V is a subset $W \subseteq V$ which contains 0_V and is closed under the addition and scalar multiplication operations of V.

Remark: In mathematics, there is generally a bit of ambiguity in the word "subset:" is a set A a subset of itself? The answer is yes - and there are good reasons for it to be thus - but we often want to talk about subsets $B \subseteq A$ which are not equal to A itself. These are called *proper subsets* (you might also hear "strict subsets"), and the word "proper" is frequently used with this connotation. For instance, a *proper subspace* $W \subset V$ is a subspace of V which is not equal to V. Often, results which are true of subsets are only "interesting" in the case of proper subsets; in some cases, a statement or construction *requires* that a subset be proper. So it is useful to have a separate term for this phenomenon.

Notation: There is also some contention about the best notation for proper subsets. The correct notation is to use the symbol $\subseteq$ to mean "is a subset of" and $\subset$ to mean "is a proper subset of" (by analogy with the symbols $\leq$ and $<$). Since this distinction is unimportant most of the time (and since it is sometimes more "careful" to leave open the possibility that a subset might not be proper), many mathematicians use the symbol $\subseteq$ every time (even when writing about subsets which are necessarily proper), and many others use the symbol $\subset$ even when writing about subsets which are not proper. So when we want to emphasize that one set is a *proper* subset of another, we sometimes have to use the unsightly symbol $B \subsetneq A$ (although it's usually enough to say "proper subset" in words and then use whichever symbol you like).

Exercise: Give geometric interpretations of:

- A proper subspace of $\mathbb{R}^1$ (as a real vector space)
- A proper subspace of $\mathbb{R}^2$
- A proper subspace of $\mathbb{R}^3$

Challenge: The complex numbers, $\mathbb{C}$, can be depicted as a plane in the same way that the real numbers can be depicted as a line. Consider the complex vector space $V = \mathbb{C}^1$, and let W be the horizontal line through the origin, i.e. the x-axis. Is W a subspace of V over the complex numbers?

Exercise: Let $V = \mathbb{R}^3$ (so an element of V can be denoted by an ordered triple (x, y, z), with $x, y, z \in \mathbb{R}$). Let $W \subset V$ be the set of triples (x, y, z) such that $3x + 2y - z = 0$. Show that W is a subspace of V.

Exercise: Let P be the vector space of polynomials with real coefficients; as discussed in class, P is a vector space over $\mathbb{R}$. Let Q denote the set of polynomials p such that $p(105) = 0$, and let S denote the set of polynomials p such that $p(0) = 1$. Show that Q is a subspace of P, but S is *not* a subspace of P.

Exercise: Let $V = \mathbb{R}^3$ as before, and let X be the set of ordered triples (x, y, z) such that $x^2 + y^2 + z^2 \leq 1$. Show that X is not a subspace of V.

4 Functions

One of the major developments in mathematics over the past century has been the realization that functions between objects are often more interesting than the objects themselves. As we'll soon see, individual vector spaces are actually rather boring - there isn't that much you can say about them, and in some sense they're mostly the same. Functions between vector spaces satisfying certain additional properties - which we will soon call "linear maps" - are, on the other hand, extremely rich. Since we'll be working quite extensively with functions this quarter, we should spend some time getting familiar with their basic properties.

Intuition: Let X and Y be sets. A *function*, or *map*, from X to Y, denoted $f : X \to Y$, is a correspondence which assigns to each element of X a single element of Y. You can think of this as a machine which "eats" an element of X and "spits out" an element of Y, and which always gives the same output for a given input.

Note that, unlike in computer science, we do not care about the "inner workings" of a function, and in some sense a function has no "inner workings". For us, a function is defined entirely by the outputs it assigns to each input. So, $f(x) = |x|$ and $g(x) = \sqrt{x^2}$ give the same function $\mathbb{R} \to \mathbb{R}$, even though they are computed in different ways.

Here is the actual definition of a function, which can be a bit opaque.

Definition: A function $f : X \to Y$ is a set of pairs (x, y), with $x \in X$, $y \in Y$, such that for each $x \in X$ there is a unique $y \in Y$ for which $(x, y) \in f$.

Exercise/Challenge: Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Explain how the "graph" of f is in fact the function f itself.

But the formal definition of function hardly matters, and as far as I know we won't make you use this definition in any proofs (although if Sihao says otherwise, you should believe him over me). Here are some properties we *will* use extensively.

Definition: Let $f, g : A \to B$ be two functions. We say $f = g$ if for all $a \in A$, $f(a) = g(a)$.

Challenge: Show that the above definition follows from the formal definition of a function.

Definition: Let A and B be sets. We say that a function $f : A \to B$ is injective if it sends each element of A to a distinct element of B; that is, if we have elements $a_1, a_2 \in A$ and $f(a_1) = f(a_2)$, then $a_1 = a_2$. We say that f is surjective if it can output every element of B; that is, if for every element $b \in B$, we can find an element $a \in A$ such that $f(a) = b$. We say that f is bijective if it is injective and surjective.

Exercise: Let A, B, C be sets, and let $f : A \to B$ and $g : B \to C$ be functions. We denote by $g \circ f$ the function $A \to C$ given by $(g \circ f)(a) = g(f(a))$; this operation on functions is called *composition*.

(a) Show that if f and g are both injective, then $g \circ f$ is injective.

(b) Show that if f and g are both surjective, then $g \circ f$ is surjective.

(c) Using parts (a) and (b), show that if f and g are both bijective, then $g \circ f$ is bijective.

Exercise: As above, let A, B, C be sets and $f : A \to B$, $g : B \to C$ be functions.

(a) Show that if $g \circ f$ is injective, then f is injective.

(b) Show that if $g \circ f$ is surjective, then g is surjective.

(c) Give an example of a case where $g \circ f$ is injective but g is not. (Hint: you can do this using finite sets)

(d) Give an example of a case where $g \circ f$ is surjective but f is not. (Hint: you can do this with finite sets as well).

Definition: Let A be a set. The *identity function* $1_A : A \to A$ is defined by $1_A(a) = a$. Sometimes this function is denoted by Id_A or I_A, and sometimes we will omit the subscript for convenience.

Exercise: Let A and B be sets and $f : A \to B$ a function. Show that $f \circ 1_A = f = 1_B \circ f$.

Definition: Let A and B be sets and $f : A \to B$ a function. A function $g : B \to A$ is called a *left inverse* to f if $g \circ f = 1_A$; it is called a *right inverse* if $f \circ g = 1_B$. If g is both a left inverse and a right inverse, it is called an *inverse* of f, sometimes denoted f^{-1}.

Exercise: Let A and B be sets, and $f : A \to B$ a function.

(a) Show that if f has a left inverse, then it is injective.

(b) Show that if f has a right inverse, then it is surjective.

(c) Conclude that if f has a right and left inverse, then it is bijective. (Note: this follows easily from parts (a) and (b)).

(d) If f is bijective, show that it has an inverse. (Note: this does *not* follow easily from parts (a)-(c); the proof will take several lines!)

Challenge: Let A and B be sets and $f : A \to B$ a function. Suppose that $g_1, g_2 : B \to A$ are both inverses (i.e. they are both left and right inverses) to f. Show that $g_1 = g_2$, that is, inverses are unique (and so we can meaningfully speak of "the inverse" of a function). As an extra challenge, show that left inverses (or right inverses) are not unique.